

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (20 marks – approximately 36 minutes)

Timberline Company Limited ("Timberline"), a custom-made wood furniture manufacturer in Lantau Island, is experiencing financial distress despite of increased sales revenue in recent years. Timberline has the following account balances for current assets and current liabilities for the financial years ended 31 December 2018 and 2019:

	<u>2018</u>	<u>2019</u>		<u>2018</u>	<u>2019</u>
	HK\$'000	HK\$'000		HK\$'000	HK\$'000
Cash	185	338	Accounts Payables	1,250	2,680
Accounts Receivables	1,560	3,980	Accruals	860	1,260
Inventory	962	3,890			
Current Assets	<u>2,707</u>	<u>8,208</u>	Current Liabilities	<u>2,110</u>	<u>3,940</u>

Timberline generated an increase in sales revenues of 25% to arrive HK\$12,450,000 with all costs and operating expenses (including interest and tax) amounted to HK\$10,680,000 in the financial year 2019. Depreciation for the year amounted to HK\$462,000, which is included in the above amount of costs and operating expenses. Capital expenditure on acquisition of plant and machinery during the year amounted to HK\$3,400,000.

Required:

- (a) Prepare an extract of the statement of cash flows showing Timberline's "cash flow from operating activities" for the financial year 2019 using indirect method. (9 marks)
- (b) Analyse the implications of the change in each component of current assets and current liabilities (other than cash) on the amount of operating cash flows calculated in Question 1(a) above. (6 marks)
- (c) Define and calculate "free cash flow". Advise the financial strategy to meet current obligation for Timberline in the financial year 2019. (5 marks)

Question 2 (20 marks – approximately 36 minutes)

- (a) Recently, you inherited HK\$600,000 cash and 65,000 shares in ICDC currently worth HK\$400,000 from your father. In his will, he specifically requests that the ICDC shares are intended to be held for long-term investment and not allowed to be sold within 20 years, and the HK\$600,000 cash must be completely invested in one of the following two stocks to form a portfolio with ICDC shares: CK Property or HK Mobile.

The table below summarises the correlation coefficients, expected annual returns, and annual standard deviations of the three stocks.

	Correlation coefficients			Expected returns	Standard deviations
	ICDC	CK Property	HK Mobile		
ICDC	1	0.8	0.3	18%	23%
CK Property	0.8	1	0.45	23%	34%
HK Mobile	0.3	0.45	1	28%	48%

Required:

- (i) Explain, with calculation, the highest expected return for the portfolio achievable under given restrictions.

(4 marks)

- (ii) Explain, with calculation, the attainable lowest risk for the portfolio under mentioned restrictions.

(6 marks)

- (b) After coming across trading investment in the stock market, you have a keen interest in learning more about the financial environment in which a firm operates.

Required:

- (i) State the ultimate goal of the financial management in a firm. Explain whether and how a financial manager's decisions can affect this goal with an example.

(3 marks)

- (ii) Explain the difference between money markets and capital markets.

(2 marks)

- (iii) Identify the relationship between initial public offerings ("IPOs") and primary market. Identify the capital market where the public trades stocks on the Hong Kong Stock Exchange.

Explain the impact of marketability on transaction costs of investors and companies trading stocks on the stock exchange.

(5 marks)

Question 3 (20 marks – approximately 36 minutes)

The Relaxing Recreation Club ("RRC") has two clubhouses. The main clubhouse is situated in Sai Kung on four and half acres of luxuriant grassland facing a picturesque and paradisiac beach and the second clubhouse is just 15 minutes' drive from the central business district and the best location on Hong Kong Island for water sports. RRC is considering a four-year renovation project to improve its facilities and equipment. You are the Chief Financial Officer of RRC and are responsible for estimating the discount rate, which is the cost of capital, used for capital budgeting appraisal. The capital structure of RRC with relevant market information is shown as below:

Common stock: 50 million shares outstanding of HK\$5 par selling for HK\$12 per share. The stock has a beta coefficient of 1.6.

Preferred stock: 8 million shares selling for HK\$94 per share, with dividend rate of 6% and face value of HK\$100.

Debt: Five years ago, RRC issued 7 million 15-year zero coupon bonds (assume compound semi-annually for bond price computation as compared with normal semi-annual coupon bonds) with par value of HK\$1,000 that are still outstanding now. The bonds are currently selling for HK\$620.

Market: The current Exchange Fund Bills yield 3% and the expected return on the market is 12%. RRC is in the 20% corporate tax bracket.

The detailed information of the initial cash outflow and cash inflows of the two mutually exclusive four-year renovation projects for evaluation is listed as below:

<u>Year</u>	<u>Project A</u>	<u>Project B</u>
	HK\$	HK\$
0	-1,280,000	-738,000
1	289,000	138,000
2	336,000	258,000
3	570,800	301,000
4	823,400	598,000

Required:

- (a) Compute the cost of equity (R_E), cost of preferred stock (R_P) and pre-tax cost of debt (R_D) for RRC. (6 marks)
- (b) Assuming that RRC is going to maintain the current capital structure, compute the weighted average cost of capital ("WACC") for project evaluation. (3 marks)
- (c) Compute the net present value ("NPV") for Project A and Project B and determine which project RRC should choose with brief explanation. (5 marks)

- (d) Compute the profitability index ("PI") for each project and determine which project RRC should choose with brief explanation. (3 marks)
- (e) Based on the answers of Questions 3(c) and 3(d), determine which project RRC should undertake for renovation with brief explanation. (3 marks)

Question 4 (20 marks – approximately 36 minutes)

Smart Sports Station ("SSS") manufactures and sells two types of sports equipment: Badminton Rackets and Tennis Rackets. Segment reporting is a useful tool to present the financial information for each business division for performance evaluation. As a Chief Operating Officer of SSS, you would like to adopt the responsibility accounting system, holding managers accountable for the performance under their control, to measure the profitability of the two segments, Badminton Rackets and Tennis Rackets. Their production information for the first quarter is extracted as below:

	<u>Badminton Rackets</u>	<u>Tennis Rackets</u>
Sales (in units)	60,400	30,800
	HK\$	HK\$
Sales price (per unit)	300	900
Variable production costs (per unit)	90	350
Traceable fixed production costs	6,200,000	9,400,000
Variable selling expenses (per unit)	35	56
Traceable fixed selling expenses	300,000	450,000
Allocated portion of the corporate expenses	3,480,000	4,600,000

Required:

- (a) Define what segment reporting is and state whether publicly held entities or privately held companies are required to undertake segment reporting. Explain why it is useful for decision making by investors and creditors. (4 marks)
- (b) Prepare a segmented statement of profit or loss with contribution margin and segment margin for the first quarter, showing columns for the company as a whole and for each product segment of Badminton Rackets and Tennis Rackets respectively. (8 marks)
- (c) State the FOUR types of responsibility centres and explain how their managers are evaluated for performance measurements. Identify which responsibility centre the Badminton Rackets segment belongs to. (8 marks)

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